SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
Student Name:	M.Manoj Kumar	**Reg. No.:**	192425114
Section / Batch:	B.Tech AI &ML	**Date:**	08-08-2026

Code of Conduct

I _____M.Manoj Kumar192425114_______ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _M.Manoj Kumar________________________

Instructions

- Read each industry scenario carefully before answering.
- Analyze the given problem from a semantic analysis perspective.
- Provide structured and logical answers with proper justification.
- Support your analysis using examples from the given scenario wherever applicable.
- Use suitable diagrams, semantic representations, logical predicates, workflows, architectures, or tables whenever necessary.
- Clearly state assumptions if additional information is required.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic interpretation of user queries, identify ambiguity in meaning representation, evaluate semantic processing limitations, and improve understanding in banking chatbot applications.	Q1
Syntax-Driven Semantic Analysis	Analyze multiple interpretations of spoken commands, evaluate parsing-related ambiguity, and assess semantic extraction in real-time voice assistant systems.	Q2
First-Order Predicate Calculus, Representing Linguistically Relevant Concepts, Syntax-Driven Semantic Analysis	Design a semantic processing architecture for healthcare reports, model relationships among entities and actions, represent linguistic concepts, and improve semantic extraction accuracy.	Q3

Q1. AI-Powered Banking Customer Support Chatbot

A banking company is developing an AI-powered customer support chatbot to understand and respond to customer queries. The chatbot uses Context-Free Grammar (CFG) for syntactic parsing. However, it faces difficulties with ambiguous sentences, subject–verb agreement, and long conversational queries. The given query is “Show me the transactions with the card from last month.”

a) Analysis of Ambiguity

The given sentence has both syntactic and semantic ambiguity. The phrase “with the card” can be interpreted in different ways. It may indicate transactions performed using a particular card, or it may refer to transactions associated with a card. The phrase “from last month” also needs to be correctly connected with the transactions.

A basic CFG can identify grammatical components such as noun phrases, verb phrases, and prepositional phrases. However, it cannot completely determine which interpretation is intended by the customer.

b) Limitations of CFG

CFG is useful for representing the basic syntactic structure of sentences, but it has several limitations in real-world banking applications.

• It mainly focuses on syntax rather than complete semantic understanding. • It may generate multiple parse trees for ambiguous sentences. • It does not naturally represent contextual information. • Handling subject–verb agreement requires additional mechanisms.

c) Proposed Improved Solution

A hybrid parsing approach using CFG, PCFG, feature structures, and Earley parsing can be implemented.

CFG can first construct the possible syntactic structures. PCFG can then assign probabilities to these structures and select the most likely interpretation.

User Query
↓
Tokenization
↓
CFG Parsing
↓
Generate Possible Parses
↓
PCFG Ranking
↓
Feature Structure Checking
↓
Semantic Interpretation
↓
Banking Response

d) Industry Benefits

The proposed approach can improve the chatbot in several ways. It reduces incorrect interpretations and provides more reliable responses to customers.

Q2. Voice Assistant System

A voice assistant system processes spoken commands using parsing techniques. The existing system uses top-down parsing but experiences backtracking problems, difficulty with incomplete or ambiguous inputs, and challenges in providing real-time responses. The example command is "Book a flight to Delhi with a window seat."

a) Analysis of Multiple Interpretations

The given command contains several important components. "Book a flight" represents the main action, "Delhi" represents the destination, and "window seat" represents the seating preference.

The semantic representation can be written as:

During parsing, the system must correctly identify the relationship between the different phrases. If the relationships are incorrectly identified, the voice assistant may produce an incorrect booking request.

b) Limitations of Top-Down Parsing

Top-down parsing begins with the start symbol and attempts to generate a suitable parse for the input. When several grammar rules can be applied, the parser may need to backtrack and try alternative possibilities.

The major limitations are:

- Excessive backtracking for ambiguous commands.
- Increased processing time for complex sentences.
- Difficulty with incomplete spoken input.
- Poor handling of unexpected conversational expressions.
- Reduced suitability for real-time applications.
- Multiple possible interpretations may delay the final response.

These limitations are important because voice assistants are expected to respond quickly to spoken commands.

c) Use of Earley Parsing

Earley parsing is suitable for handling ambiguous and complex sentences. It maintains partial parsing information while processing the input. Instead of repeatedly restarting the parsing process, it keeps track of possible structures.

d) Comparison of Top-Down and Earley Parsing

Top-down parsing is relatively simple and works well with simple and predictable grammatical structures. However, its performance can decrease when the input is ambiguous or requires extensive backtracking.

Earley parsing is more flexible and can handle ambiguous, incomplete, and complex inputs. It maintains different possible states during parsing and can therefore be more suitable for conversational applications.

For a real-time voice assistant, Earley parsing provides better support for ambiguity and partial input handling. Hence, it is a suitable alternative to basic top-down parsing for this application.

Q3. Healthcare Medical Report Analysis

A healthcare company is developing an NLP system to analyze patient medical reports and automatically extract important information such as diagnosis, prescribed treatment, and follow-up actions. The system must handle complex medical sentences, structural ambiguity, agreement, relationships between medical entities and actions, and real-time processing. The example sentence is "The doctor who reviewed the patient last week recommends starting medication and scheduling a follow-up visit in Chennai."

a) Complete NLP Architecture

A suitable architecture for the healthcare system is:

Medical ↓ Report

Text Preprocessing
↓
Tokenization
↓
CFG-Based Syntactic Parsing
↓
Feature Structure Analysis
↓
PCFG-Based Ambiguity Resolution
↓
Sub-Categorization ↓ Analysis

Semantic Extraction
↓
Structured Medical Information

CFG provides the basic syntactic structure of the medical sentence. PCFG helps select the most probable interpretation when more than one parse is possible. Feature structures represent linguistic properties and help maintain agreement. Efficient parsing techniques support faster processing of complex medical reports.

b) Step-by-Step Processing

Step 1: Input the medical report into the NLP system.

Step 2: Preprocess the text by separating words, phrases, and relevant linguistic units.

Step 3: Apply CFG to determine the syntactic structure of the sentence.

Step 4: Use feature structures to identify grammatical relationships and agreement.

Step 5: Apply PCFG to resolve possible structural ambiguities using probabilities.

Step 6: Identify important medical entities such as doctor, patient, medication, visit, and location.

Step 7: Apply sub-categorization frames to determine relationships between medical actions and their arguments.

Step 8: Convert the extracted information into structured output for further medical processing.

c) Structured Output

For the given sentence, the system can produce the following information:

Doctor: The doctor

Previous action: Reviewed the patient

Time reference: Last week

Treatment action: Starting medication

Follow-up action: Scheduling a follow-up visit

Location: Chennai

The sentence does not explicitly provide a diagnosis, so the system should not generate a diagnosis that is not present in the input.

d) Sub-Categorization Frames

Sub-categorization helps identify the arguments required by medical verbs. For example:

recommend(Doctor, Action)

Here, the doctor is the entity performing the action, while the recommended action is the starting of medication and scheduling of a follow-up visit.

This approach helps the system understand relationships instead of simply identifying individual words. It is particularly useful for medical reports where the relationship between a patient, doctor, treatment, and follow-up action is important.

e) Real-Time Processing

For hospital applications, the system should process medical reports quickly. Efficient parsing techniques can reduce unnecessary computation. PCFG can be applied for resolving important ambiguities, while feature structures can provide grammatical consistency. Processing can also be divided into smaller stages so that information can be extracted without waiting for the entire hospital database to be processed.

f) Accuracy and Scalability

Accuracy can be improved by combining syntactic, probabilistic, and semantic techniques rather than depending only on CFG. Medical-specific vocabulary and sub-categorization rules can help identify relationships between medical actions and entities.

For scalability, the system should use efficient parsing methods and modular processing. This allows the same architecture to process increasing numbers of medical reports while maintaining reasonable response time. The proposed combination of CFG, PCFG, feature structures, and efficient parsing addresses the major requirements of the healthcare application.

Conclusion

CFG provides the basic syntactic foundation, while PCFG helps resolve ambiguity, feature structures handle linguistic relationships, and efficient parsing improves processing speed. Combining these techniques provides a stronger semantic processing framework for banking chatbots, voice assistants, and healthcare NLP systems.

Rubrics for Calculation

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Q. No. / Criteria Level	**Understanding of Industry Problem**	**Application of Engineering Concepts**	**Solution Design**	**Use of Tools**	**Analysis**	**Professionalism**	**Sub. Total**	**Total %**

1								
2								
3								

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ________________ Date: ____________________	Signature: ________________ Date: ____________________	Signature: ________________ Date: ____________________